

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTHM505 - PROJECT MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

1 Which of the following is not a component of project management?

- a) Material management
- b) Machine design
- c) Manpower management
- d) Network scheduling

In a Bar Chart, activities are represented:

- a) As nodes
- b) As arrows
- c) As bars on a time scale
- d) As milestones

3 Which of the following is used to calculate the earliest and latest start/finish times in a network?

- a) Bar chart
- b) Fulkerson's Rule
- c) Critical Path Method (CPM)
- d) PERT

4 A project has three time estimates for an activity: Optimistic = 4 days, Most Likely = 6 days, and Pessimistic = 10 days. What is the expected time using PERT?

- a) 6.0 days
- b) 6.33 days
- c) 7.0 days
- d) 5.5 days

5 Crashing a project involves:

- a) Reducing project quality
- b) Reducing project duration with least additional cost
- c) Increasing resources without any cost concern
- d) Adding more activities to the network

- 6 In resource allocation, if resources are limited, the project schedule is adjusted to:
- Decrease cost
 - Eliminate the critical path
 - Remove dependencies
 - Delay non-critical activities
- 7 The standard deviation for a PERT activity with time estimates $O = 3$, $M = 6$, $P = 15$ is:
- 2
 - 1.5
 - 3
 - 4
- 8 Which distribution is used in PERT to model time estimates?
- Normal distribution
 - Poisson distribution
 - Beta distribution
 - Exponential distribution
- 9 If the total project duration is 40 days with a standard deviation of 4 days, what is the probability the project will be completed within 44 days ($Z = 1$)?
- 68%
 - 84%
 - 95%
 - 50%
- 10 Which of the following cost types is considered a one-time expenditure?
- Operating cost
 - Fixed cost
 - Capital cost
 - Variable cost
- 11 The present worth of ₹10,000 to be received 3 years from now at an interest rate of 10% per annum is closest to:
- ₹7,513
 - ₹8,000
 - ₹7,200
 - ₹6,500
- 12 Which of the following is not part of Deming's philosophy of Total Quality Management?
- Eliminate slogans and targets
 - Cease dependence on mass inspection
 - Maximize defects to learn from them
 - Drive out fear in the organization
- Q2. Solve the following.
- Explain Bar chart and Milestone chart in detail. 6
 - What are the Stages of Project Management 6
- Q3. Solve the following.
- Differentiate between CPM and PERT. 6

B) Consider the following table summarizing details for project.

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Activity		A	B	C	D	E	F	G	H	I	J
Predecessors		-	-	-	A	B	C	C	E,F	D	H,G
Duration (week)	To	5	1	1	1	1	1	2	4	2	2
	T _m	6	3	4	2	2	5	2	4	5	2
	T _p	6	3	4	2	3	5	3	5	5	3

Construct the project network diagram. Find the expected duration and variance of each activity. Find the critical path and expected project completion of time.

Q4. Solve Any Two of the following.

A) Define and explain Optimistic time, Pessimistic time, and Most likely time.

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B) Write a short note on Gantt charts

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C) Draw the network and find the project duration & Critical path for the following

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Activity (i-j)	t _o (Optimistic Time)	t _L (Most likely Time)	t _p (Pessimistic Time)
1-2	2	5	8
2-3	8	11	20
3-4	0	0	0
2-4	4	7	16
2-5	4	9	20
4-6	7	10	13
5-6	3		17
3-7	3		13
6-7	2		10
7-8	2		6

Q5. Solve Any Two of the following.

A) A manufacturing company purchased a CNC Machine at an initial cost of ₹60 lakhs. The machine incurs annual maintenance and electricity charges of ₹3 lakhs. The machine generates annual revenue of ₹12 lakhs through production activities. After 8 years of use, the machine has a salvage value of ₹6 lakhs.

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i) Draw the Cash Flow Diagram for the transaction.

ii) Calculate the total expenses, total earnings, and net profit over the 8-year period.

B) Explain Demand and Supply curve, Inflation and Cash flow diagram.

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C) What is the Crashing Heuristic?

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Q6. Solve Any Two of the following.

A) Explain in detail the concept of Quality Control.

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B) Draw the network and calculate TE & TL for all activities for the following.

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Activity (i-j)	1-2	1-3	1-4	2-5	2-7	3-4	3-6	4-5	5-6	5-7	6-7	7-8
Duration (t _{ij}) Weeks	6	7	8	7	8	4	4	9	5	6	6	9

C) Explain the process for Post-Project Analysis.

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